

Electronics Workshop: *LTSpice*

Project Overview

Welcome to the **Purdue MIND Electronics Onboarding Project!** In this project, you will design and simulate an analog filter circuit using **LTSpice**, then optionally translate your circuit into a PCB design using **KiCad**.

Throughout this project, you will gain experience with **circuit simulation**, **frequency response analysis**, **schematic design**, and **PCB layout**.

Estimated time: 2-3 hours

LTSpice Installation

- 1) Navigate to <https://www.analog.com/en/resources/design-tools-and-calculators/ltspice-simulator.html>
- 2) Scroll down to the “**Download LTSpice**” section and install the software based on your operating system

The screenshot shows the LTSpice download page on the Analog Devices website. On the left is a navigation menu with links: EE-Sim, LTSpice (highlighted), Power Management, RF & Synthesis, and Cybersecurity. The main content area features a description of LTSpice as a powerful, fast, and free SPICE simulator software. It includes a section titled 'Download LTSpice' with instructions to download for various operating systems. Two download buttons are highlighted with a red box: 'Download for Windows 10 64-bit and forward' (Version 24.1.9) and 'Download for MacOS 10.15 and forward' (Version 17.2.4). Below these are links for older versions: 'Download for Windows XP (End of Support)', 'Download for MacOS 10.9 (End of Support)', and 'Download LTSpice XVII for Windows (End of Support)'. At the bottom, there are sections for 'Get Support' and 'Find Demo Circuits'. A vertical 'Site Feedback' button is on the right edge.

EE-Sim
LTSpice
Power Management
RF & Synthesis
Cybersecurity

LTSpice® is a powerful, fast, and free SPICE simulator software, schematic capture and waveform viewer with enhancements and models for improving the simulation of analog circuits. Its graphical schematic capture interface allows you to probe schematics and produce simulation results, which can be explored further through the built-in waveform viewer.

Learn how to use LTSpice with [our tutorials below](#) or dive deeper with our selection of helpful tips and articles. You can also browse our library of macromodels and demo circuits for select Analog Devices products.

LTSpice's enhancements and models improve the simulation of analog circuits when compared to other SPICE solutions. Download LTSpice below to see for yourself!

Download LTSpice

Download our LTSpice simulation software for the following operating systems:
Date models updated - Jul 31 2025

Download for Windows 10 64-bit and forward Version 24.1.9
Download for MacOS 10.15 and forward Version 17.2.4

[Download for Windows XP \(End of Support\)](#)
[Download for MacOS 10.9 \(End of Support\)](#)
[Download LTSpice XVII for Windows \(End of Support\)](#)

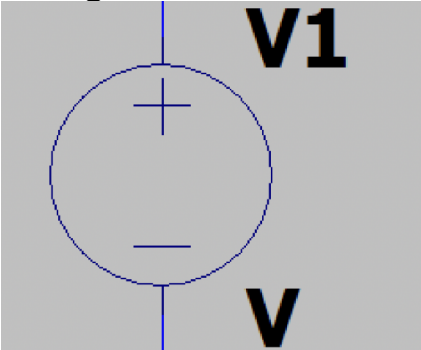
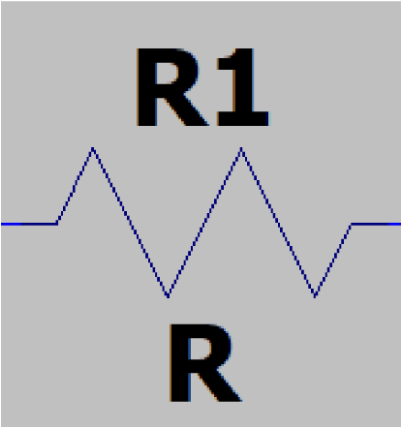
Get Support
Visit the LTSpice Forum in EngineerZone to ask questions (or answer them) or add comments about LTSpice.

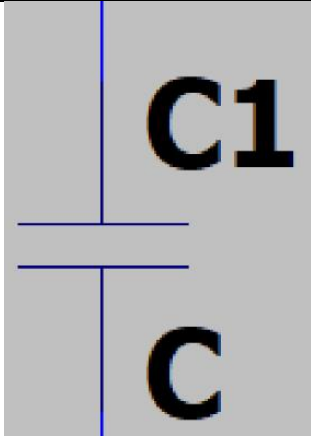
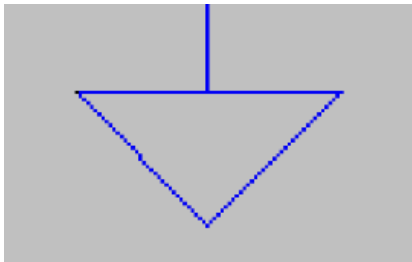
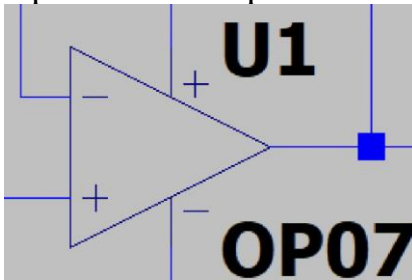
Find Demo Circuits
Explore the entire library of ready-to-run LTSpice demonstration circuits.

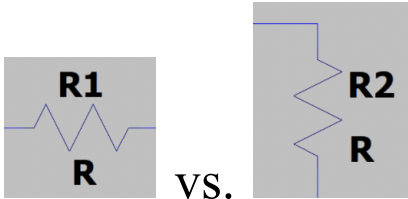
Site Feedback

Keyboard Shortcuts: Circuit Components

Refer to this as you are building your schematic:

Hit this key:	Outcome:	What the component does:
“V”	Voltage Source 	Provides the electrical voltage that powers the circuit.
“R”	Resistor 	Limits the amount of current flowing through the circuit.
“C”	Capacitor	Stores electrical energy and releases it back into the circuit.

		
"W"	Wire 	Connects components together, allowing electrical signals to travel between them.
"G"	Ground 	Provides the circuit's 0 V reference point, which other voltages are measured relative to.
"P" - The "P" key will open a menu with other components - The part labeled "OP07" on the schematic of this project is an operational amplifier. You can search for it by typing "op" then	Operational Amplifier 	Op-amp: Amplifies the difference between two input voltages to produce an output voltage.

clicking the first option		
“Ctrl-R”	Rotates parts 	

Confused on what every component does? Check out some of these resources.
 Feel free to skim!

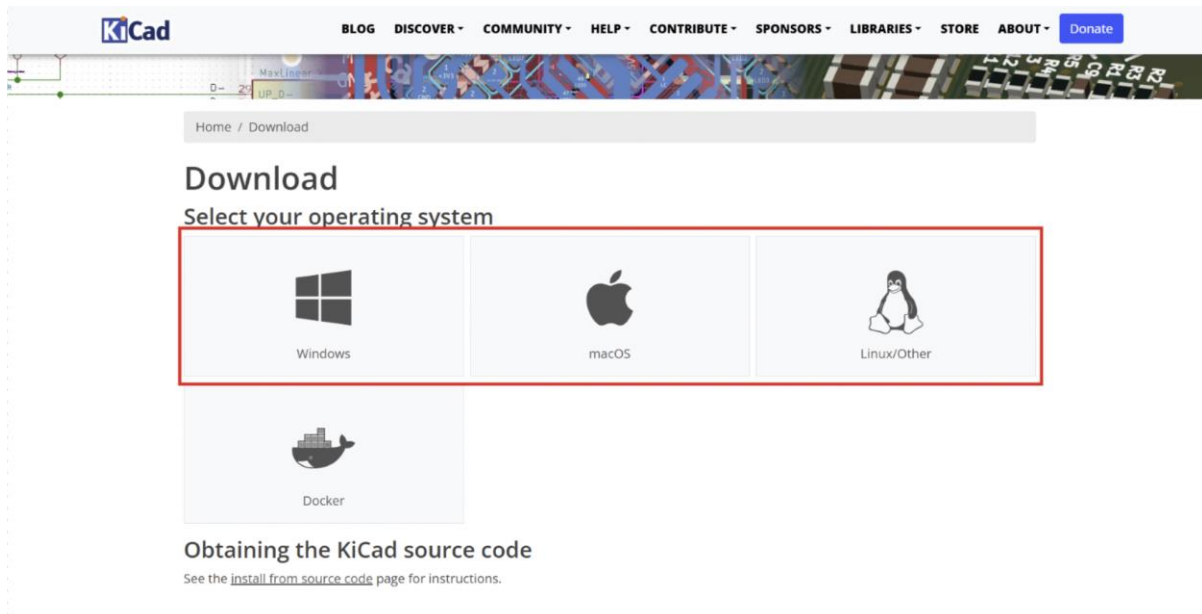
- Videos
 - [All Electronic Components Explained In a SINGLE VIDEO.](#)
 - [Schematic Diagrams & Symbols, Electrical Circuits - Resistors, Capacitors, Inductors, Diodes, & LEDs](#)
 - [DC Resistors & Batteries](#)
 - [Voltage, Electric Energy, and Capacitors](#)
- Readings
 - [How Electronic Components Work](#)
 - [A Deep Dive into 14 Essential Electrical Components](#)

KiCad Installation: *Optional*

Install after completion of the LTSpice project if you wish to complete the optional KiCad section.

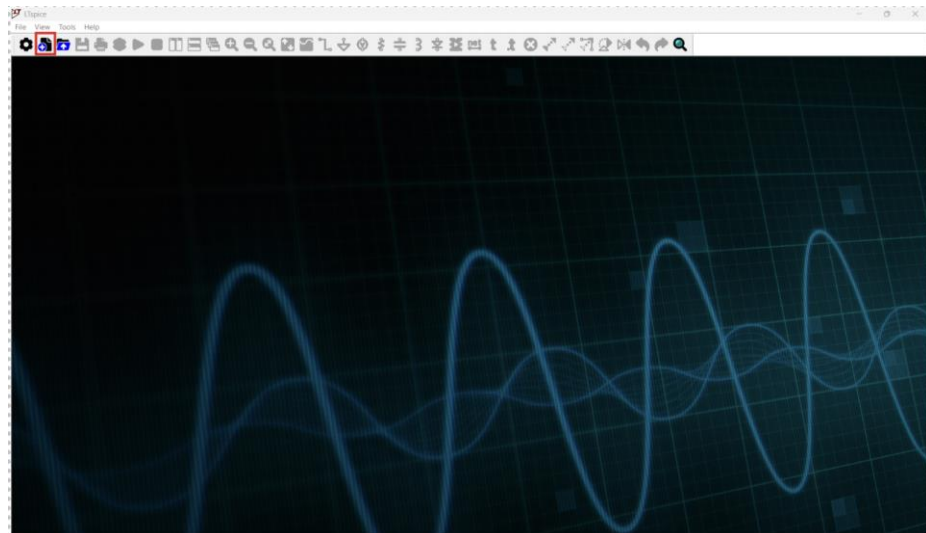
- 1) Navigate to <https://www.kicad.org/download/>

- 2) Click one of the options under “Download” to install the software based on your operating system

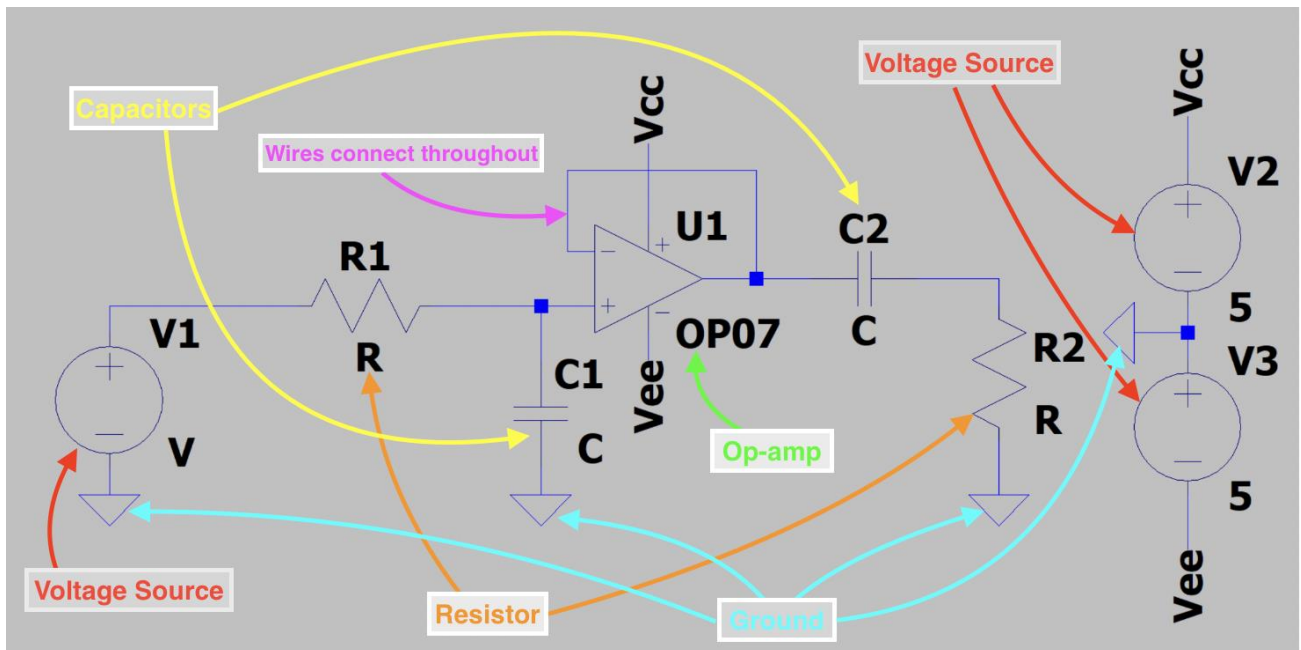
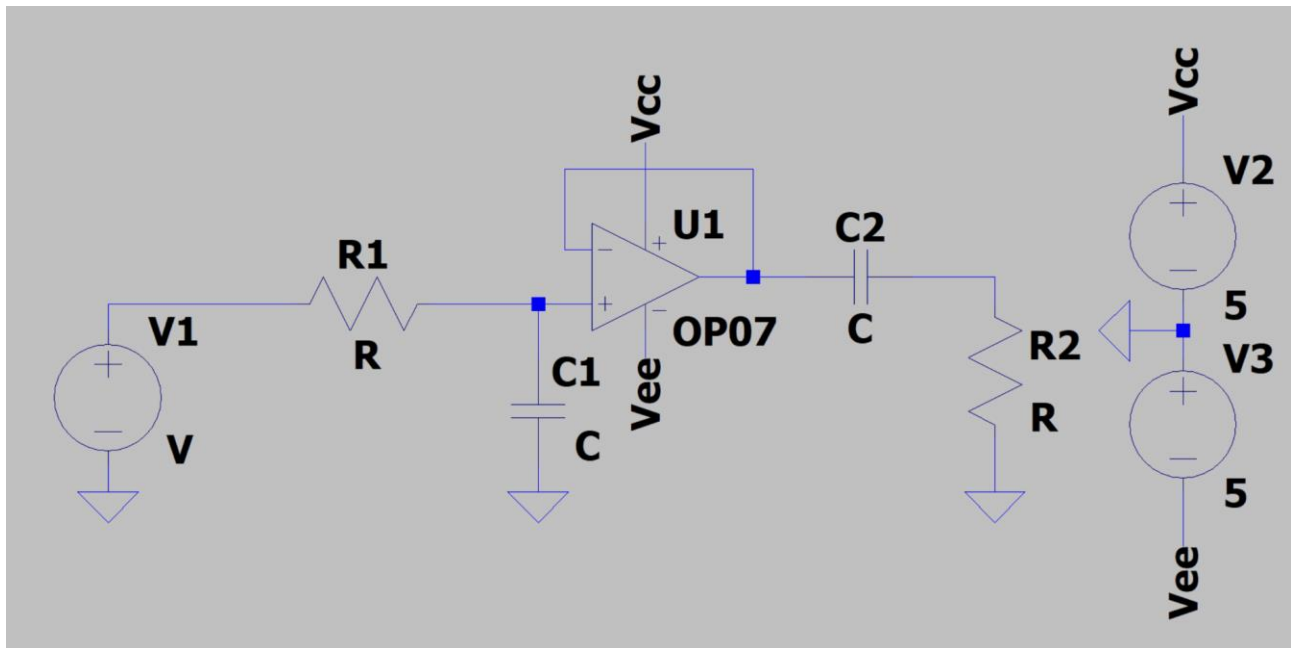


Part 1: *Building an LTSpice Schematic*

- 1) Open LTSpice on your computer
- 2) Create a blank, new schematic



3) Build the following schematic! Refer to the “Keyboard Shortcuts” at the top of this document.



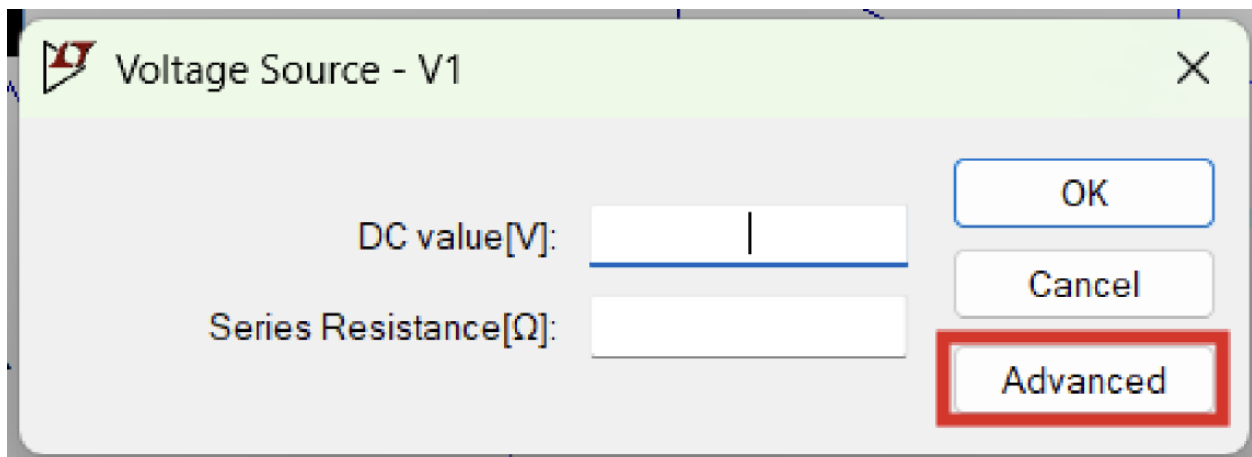
You just built a **band-pass filter**.

- A band-pass filter is composed of a high-pass and a low-pass filter connected together.

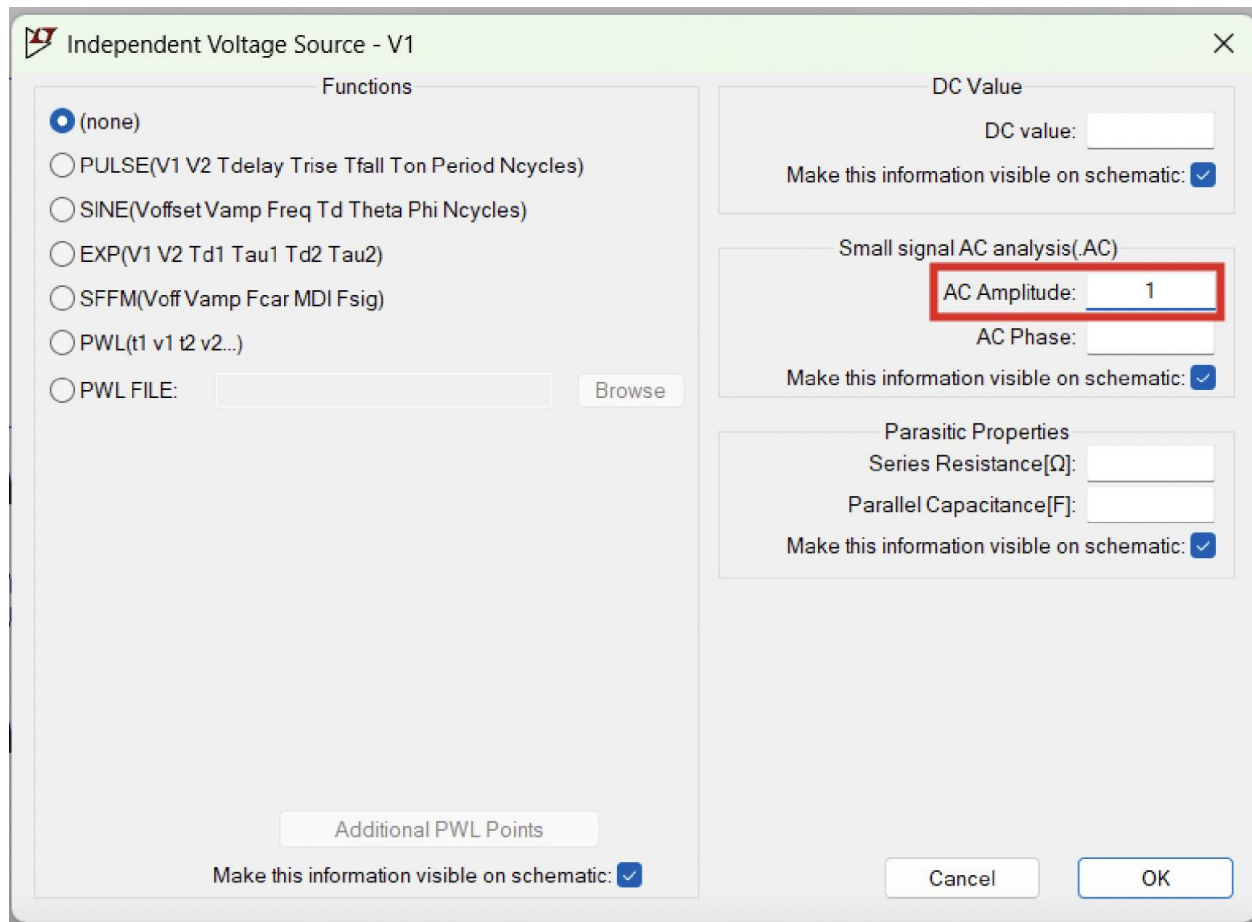
- A **high-pass filter** allows frequencies **above** its cutoff frequency to pass while attenuating lower frequencies.
- A **low-pass filter** allows frequencies **below** its cutoff frequency to pass while attenuating higher frequencies.
- In the schematic above, R1 and C1 form a low pass filter and C2 and R2 form a high pass filter.
 - The operational amplifier U1 is used to prevent the low pass filter from affecting the high-pass filter and vice versa

The arrangement of the resistor and capacitor determines whether the circuit acts as a high-pass or low-pass filter (which you just did). Their component values determine the cutoff frequency (which you will determine in the next step!).

- 4) Right-click on a voltage source. A menu should appear asking for the DC value and the series resistance.
 - a. Click on the advanced option



- 5) Enter "1" for the AC Amplitude. Then click "OK".



- 6) Press the “esc” button.
- 7) You will now **calculate to determine a resistor and capacitor combination** that achieves a cutoff frequency of 250 Hz (high-pass filter) and 5 kHz (low-pass filter).
 - a. Remember, your cutoff frequency is the point where each filter begins attenuating frequencies, which is why the high-pass filter attenuates the lower frequency bound and vice versa.

For an RC high-pass/low-pass filter, the cutoff frequency is calculated using:

$$f_c = \frac{1}{2\pi RC}$$

R = Resistance (Ω)

C = Capacitance (F)

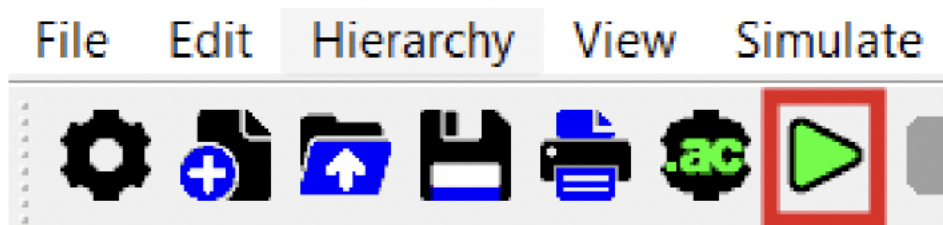
Using this equation, determine an appropriate resistor and capacitor combination for each filter. Determine R and C for both bounds.

Note: There is more than one correct combination for R and C. Different resistor and capacitor values can produce the same cutoff frequency as long as their product satisfies the equation.

HOWEVER, component values should still be reasonable. A general range for R values should fall between **1 k Ω - 100 k Ω** . A general range for C values should fall between **1 nF – 1 μ F**. These ranges represent practical values commonly used in RC filter circuits and will help you avoid unnecessarily extreme component values.

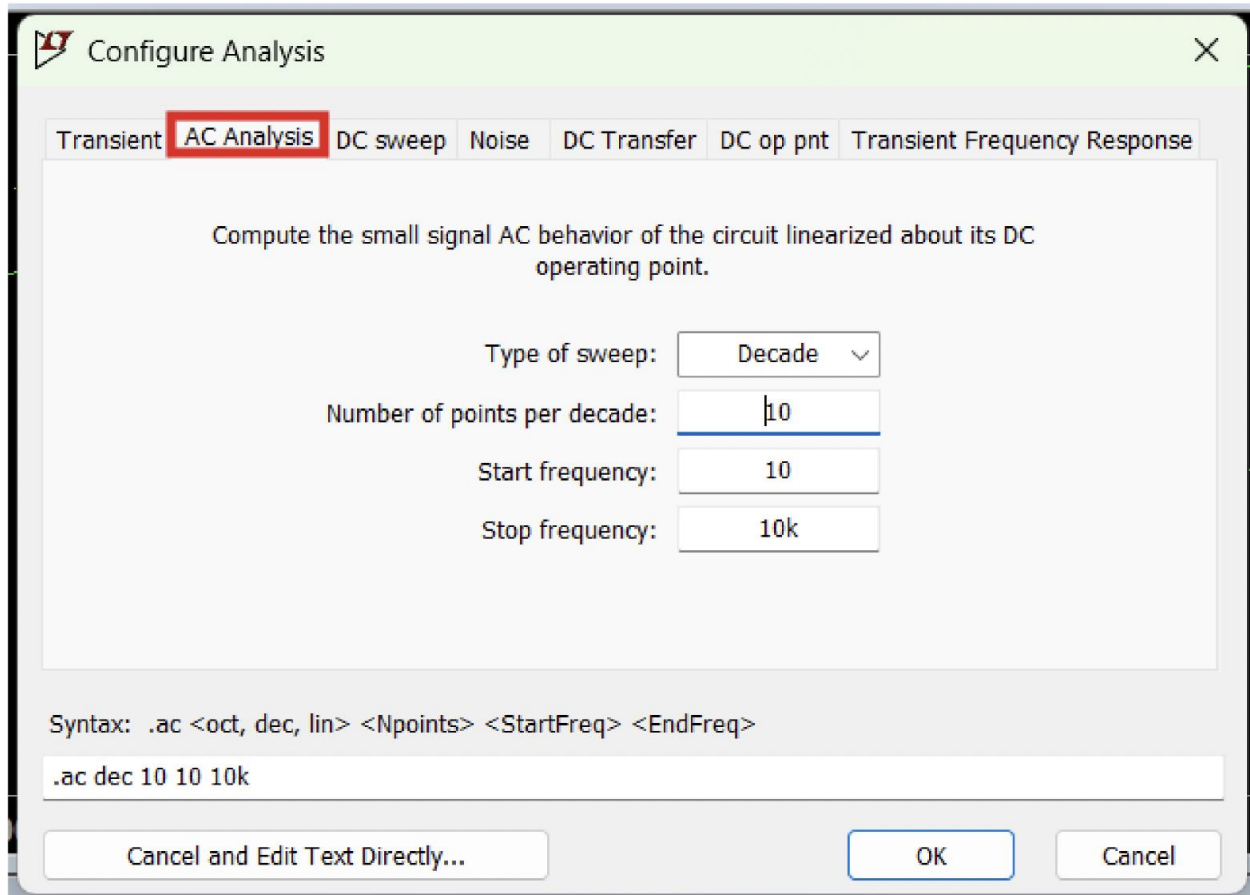
- 8) Input the values you calculated in the last step into your schematic.
 - a. Ex: If inputting your R1 value, right-click on R1 (NOT R), and enter the new value
 - b. Remember, the values you calculated for the cut-off frequency of 5kHz should go in the low pass filter (R1 and C1) and the vice versa

- 9) After you finish, click the stimulate button in the top toolbar.



- 10) Click on “AC Analysis” and input the following arguments. Then click “OK”.
 - a. **For Mac:**
 - i. Right-click any blank space on your schematic. Hover over “Draft” and click on “SPICE Directive”.
 - ii. Simply type out the argument and press “OK”.

iii. Place the text anywhere that is blank on your schematic.



The 'Configure Analysis' dialog box is shown with the 'AC Analysis' tab selected. The dialog contains a description of the analysis, configuration parameters, a syntax guide, and a command input field.

Configure Analysis

Transient **AC Analysis** DC sweep Noise DC Transfer DC op pnt Transient Frequency Response

Compute the small signal AC behavior of the circuit linearized about its DC operating point.

Type of sweep: Decade ▾

Number of points per decade: 10

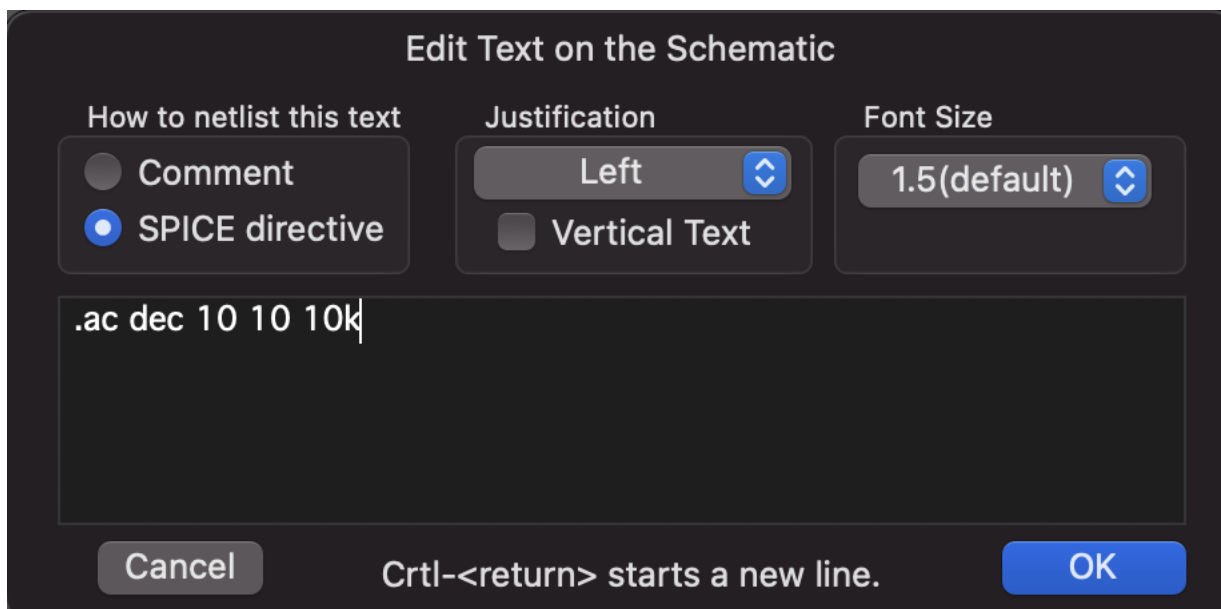
Start frequency: 10

Stop frequency: 10k

Syntax: .ac <oct, dec, lin> <Npoints> <StartFreq> <EndFreq>

.ac dec 10 10 10k

Cancel and Edit Text Directly... OK Cancel



The 'Edit Text on the Schematic' dialog box is shown with options for netlisting, justification, and font size. It includes a text input field and buttons for Cancel and OK.

Edit Text on the Schematic

How to netlist this text

☐ Comment

☒ SPICE directive

Justification

Left ▾

☐ Vertical Text

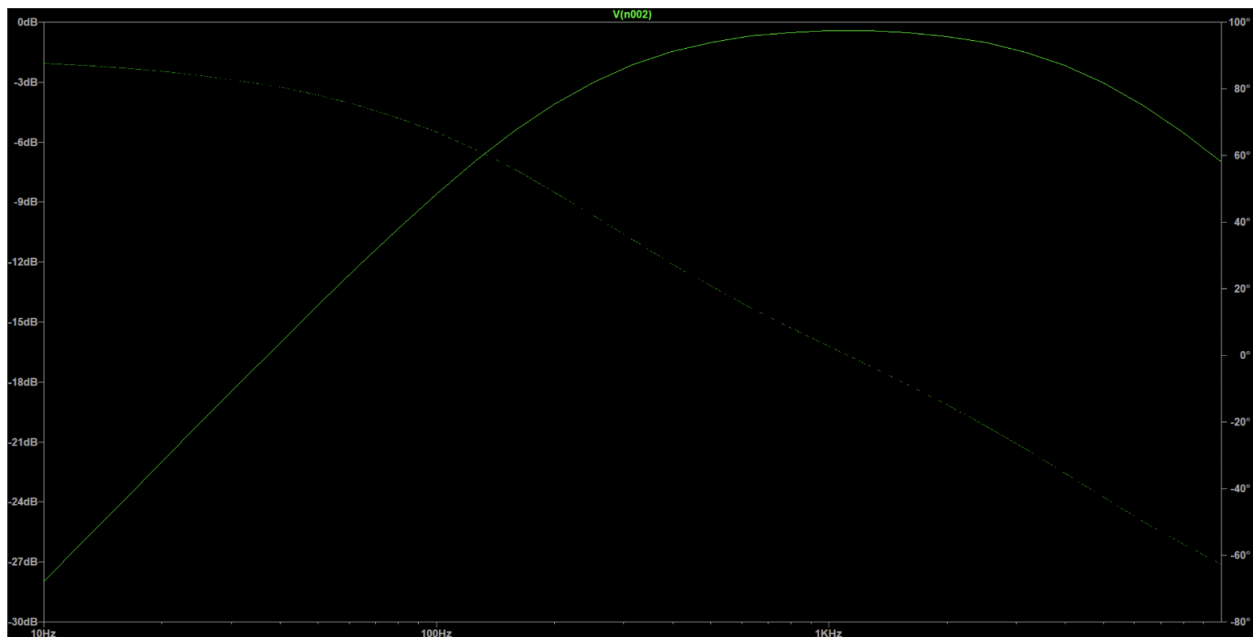
Font Size

1.5(default) ▾

.ac dec 10 10 10k

Cancel Ctrl-<return> starts a new line. OK

11) A graph like the one below should be generated:



- ☐ Verify that your cut-off frequencies are correct by clicking on the graph then pressing the "C" key
- ☐ Once you have crosshairs on the graph you can use the arrow key to scroll through the different frequencies
- ☐ **At 250Hz and 5kHz the dB metric should be about -3**
- ☐ If you don't achieve this result, verify your calculations and schematic
- ☐ If you still need help, call over a member of MIND leadership to look at your circuit

Congrats, you have completed your circuit design and simulation in LTSpice!

Find a member of MIND Leadership and get checked off!

Optional Part 2: *PCB Design in KiCad*

PDF Tutorial:

[KiCad Tutorial](#)

Video Tutorial:

[KiCad Video Tutorial.mp4](#)

Once completed, find a member of MIND Leadership and get checked off!